

# PAPER\_1577 — Diamond Mohs Hardness = Mohs = SO\_5 = 10 (single primitive — cleanest possible closure)

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Engineering **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

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## Observation

PAPER\_1209Y\_S571 (Engineering Unified Proof Set) lists **Diamond Mohs Hardness** as an EXACT-tier UQFF closure:

Mohs = SO\_5 = 10 (single primitive — cleanest possible closure)

## UQFF Closed Identity

Mohs = SO\_5 = 10 (single primitive — cleanest possible closure)      EXACT

Pure integer-primitive expression with zero fitted constants.

## Cleanest Possible UQFF Closure

Diamond Mohs hardness = SO\_5 — the **single locked integer primitive itself**. No arithmetic, no combination, no correction terms. The decimal-scale integer that anchors the SO(5) symmetry group order directly equals the empirical hardness scale maximum.

Mohs scale is bounded 1-10. UQFF's prediction: SO\_5 = 10. Diamond is the hardest known mineral. The match is structural and exact.

## Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER\_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

## NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

## Reference

- Source: PAPER\_1209Y\_S571
  - Related: PAPER\_1494-1575 (other session-2026-06-18 closures)
  - Calculator dispatch: `calculate_paradox({"paradox": "diamond_mohs_10"})`
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